

SERVICE RUNBOOK

Payments Gateway

service: payments-gateway

TIER 1

ON-CALL

1 Service Overview

`payments-gateway` handles checkout transaction processing, payment authorization, webhook handling from external payment processors, and currency conversion for international orders.

IMPACT This is a **tier-1 service** — downtime directly blocks revenue. Treat any sustained checkout degradation as a live incident.

2 Common Failure Patterns

The most frequent failure mode is **connection pool exhaustion under sustained load**, surfacing as `context deadline exceeded` errors and 504s on the checkout endpoint. This has recurred multiple times historically and should be treated as a **capacity issue** if it recurs more than twice in a quarter — not just a transient blip.

Other known failure modes include SSL certificate expiration on the public webhook endpoint (check auto-renewal service account token validity if this occurs), stale exchange rate cache causing currency mismatches on international orders, and race conditions in the order confirmation handler causing duplicate charges under concurrent webhook retries.

SYMPTOM	LIKELY CAUSE / FIRST CHECK
504 / <code>context deadline exceeded</code>	Connection pool exhaustion — check pool metrics, consider restart
SSL handshake failure on webhook endpoint	Certificate expiration — check renewal service account token

SYMPTOM	LIKELY CAUSE / FIRST CHECK
Wrong currency charged	Stale exchange rate cache — check rate update job logs
Duplicate charges reported	Race condition in webhook dedup — verify idempotency key constraint

CRITICAL / FINANCIAL IMPACT

DEGRADATION

3 Diagnostic Steps

- 1

Check `payments-gateway` pod logs for the exact error signature.
- 2

Check the **connection pool utilization dashboard** — if near capacity, this is likely a pool exhaustion event.
- 3

Check third-party **fraud detection API** and **payment processor status pages** for external outages.
- 4

Cross-reference against **prior incidents** for this service before assuming a novel cause.

4 Escalation Criteria

PAGE THE ON-CALL LEAD IMMEDIATELY IF:

- Checkout error rate exceeds **5% for more than 5 minutes**.
- Any **duplicate-charge pattern** is detected (financial / trust impact).
- A **connection pool exhaustion** event recurs for the **third time within a 90-day window**.